

Random variables

A random variable is one whose possible values are the numerical results of an on-demand real-world event. It is a rule that allocates a numerical value to each outcome in the sample space of a random experiment. In other words, a random variable is a numeric quantity whose value depends on the outcome of a random process. It is the key object that is explored in probability theory and statistics.

Types of random variables

Discrete random variable: This kind of random variable takes on a finite or countably infinite set of values. Examples: number of students in a class, number of emails per day, roll of a die.

Continuous random variable: This kind of random variable takes on uncountably infinite values in an interval. Examples: height of a person, time between arrivals, temperature.

Functions of random variables

A **probability mass function (pmf)** is a function that is denoted by $f(x)$ and describes the probability distribution of a discrete random variable $X(x)$ for several inputs taken from the sample space S . It assigns probabilities to each possible value that the variable can take. The **pmf** gives the probability of each specific value.

Properties of probability mass function:

- ✓ $f(x) > 0; x \in S$
- ✓ $\sum_{x \in S} f(x) = 1$
- ✓ $P(X \in A) = \sum_{x \in A} f(x); A \subset S$

A **probability density function (pdf)** is a function that is denoted by $f(x)$ and describes the likelihood of a continuous random variable $X(x)$ taking on values within a particular interval within the sample space S . The **pdf** itself is not a probability at a single point; it describes density. Probabilities come from integrating the density over an interval.

Properties of the probability density function:

- ✓ $f(x) > 0; x \in S$
- ✓ $\int_S f(x)dx = 1$
- ✓ $P(a \leq X \leq b) = \int_a^b f(x)dx; (a, b) \subseteq S$

Those probability functions explain the nature of a random variable and how the mean and standard deviation are distributed. Probability functions are used to create a database or statistics, which are often used in science to represent the real-valued variables, whose distribution data are not known.

The **cumulative distribution function (cdf)** of a random variable is a function that gives the probability that the variable takes a value less than or equal to an assigned value.

Let us assume $f(x)$ is the **pmf** or **pdf** of a **random variable** $X(x)$. Then, the **cdf** is denoted by $F(x)$ and is defined as

$$\begin{aligned} \text{➤ } F(x) = P(X \leq x_t) &= \begin{cases} 0; x < x_{\min} \\ \sum_{x=x_{\min}}^{x_t} f(x); x_{\min} \leq x < x_{\max} \text{ (discrete)} \\ 1; x \geq x_{\max} \end{cases} \\ \text{➤ } F(x) = P(X \leq x_t) &= \begin{cases} 0; x < x_{\min} \\ \int_{x_{\min}}^{x_t} f(x) dx; x_{\min} \leq x < x_{\max} \text{ (continuous)} \\ 1; x \geq x_{\max} \end{cases} \end{aligned}$$

Mathematical expectations of random variables

The **mathematical expectation** (also called the expectation or expected value) of a random variable is a measure of its center or average value. It is the long-run average outcome if the experiment were repeated many times.

⇒ Consider $u = u(X)$ as a function of a **discrete random variable** X with the **pmf** $f(x)$, then the **mathematical expectation** is defined as $E[u(X)] = \sum_{x \in S} u(x)f(x)$.

⇒ Consider $u = u(X)$ as a function of a **continuous random variable** X with the **pdf** $f(x)$, then the **mathematical expectation** is defined as $E[u(X)] = \int_{-\infty}^{\infty} u(x)f(x) dx$.

⇒ For $Y = X^r$, we have

$$\begin{aligned} \text{➤ } E[Y] &= E[X^r] = \sum_{x \in S} x^r f(x) \text{ (discrete)} \\ \text{➤ } E[Y] &= E[X^r] = \int_{-\infty}^{\infty} x^r f(x) dx \text{ (continuous)} \end{aligned}$$

- ❖ If the given function is renamed, or a new variable is declared with the same **pmf**, the desired mathematical expectation will not be changed.
- ❖ If the variable and/or corresponding **pmf** is redefined, the mathematical expectation will be changed.
- ❖ For a discrete random variable $P(X \geq a) \neq P(X > a)$ and $P(X \leq a) \neq P(X < a)$.
- ❖ For a continuous random variable $P(X \geq a) = P(X > a)$ and $P(X \leq a) = P(X < a)$.

Properties and applications of mathematical expectations

- ✓ $E[c] = c$, where c is a constant
- ✓ $E[cu(X)] = cE[u(X)]$
- ✓ $E[u_1(X) \pm u_2(X)] = E[u_1(X)] \pm E[u_2(X)]$
- ✓ $E[c_1u_1(X) \pm c_2u_2(X) \pm c_3] = c_1E[u_1(X)] \pm c_2E[u_2(X)] \pm c_3$

- ⇒ Mean of any distribution $\mu = E[X] = \sum_{x \in S} xf(x)$ (discrete)
- ⇒ Mean of any distribution $\mu = E[X] = \int_{-\infty}^{\infty} xf(x) dx$ (continuous)
 - $E[X - \mu] = E[X] - E[\mu] = \mu - \mu = 0$
- ⇒ Variance of any distribution $\sigma^2 = V(X) = E[(X - \mu)^2] = \sum_{x \in S} (x - \mu)^2 f(x)$ (discrete)
- ⇒ Variance of any distribution $\sigma^2 = V(X) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$ (continuous)
 - Shortcut formula $\sigma^2 = E[X^2] - \mu^2 = E[X^2] - \{E[X]\}^2$
 - If $E[X] = \mu = 0$, $\sigma^2 = E[X^2]$
- ⇒ Standard deviation $SD = \sigma = +\sqrt{V(x)} = \sqrt{E[X^2] - \{E[X]\}^2}$
- ✓ $V(u(X)) \geq 0$
- ✓ $V(c) = 0$, where c is a constant
- ✓ $V[cu(X)] = c^2 V[u(X)]$
- ✓ $V[u_1(X) \pm u_2(X)] = V[u_1(X)] + V[u_2(X)] \pm Cov[u_1(X), u_2(X)]$
 $= V[u_1(X)] + V[u_2(X)]$ [Considering $u_1(X)$ and $u_2(X)$ are uncorrelated]
- ✓ $V[c_1 u_1(X) \pm c_2 u_2(X) \pm c_3] = c_1^2 V[u_1(X)] + c_2^2 V[u_2(X)]$

Moment generating functions and their Applications

The **moment generating function (mgf)** is a powerful tool in probability theory that, when it exists, uniquely determines all the moments (means, variances, and higher-order deviations) of a random variable $X(x)$ in a single function and greatly simplifies many proofs and estimations of independent random variables. It's a useful tool for analyzing distributions and proving convergence results.

- ⇒ **The moment generating function (mgf)** is defined as $M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x)$ (discrete)
 - $M(0) = E[1] = \sum_{x \in S} f(x) = 1$
 - $M^r(t) = \sum_{x \in S} x^r e^{tx} f(x)$
 - $M^r(0) = \sum_{x \in S} x^r f(x) = E[X^r]$
- ⇒ **The moment generating function (mgf)** is defined as $M(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx$ (continuous)
 - $M(0) = E[1] = \int_{-\infty}^{\infty} f(x) dx = 1$
 - $M^r(t) = \int_{-\infty}^{\infty} x^r e^{tx} f(x) dx$
 - $M^r(0) = \int_{-\infty}^{\infty} x^r f(x) dx = E[X^r]$
- ✓ Mean $\mu = E[X] = M'(0)$
- ✓ Variance $\sigma^2 = E[X^2] - \{E[X]\}^2 = M''(0) - \{M'(0)\}^2$
- ✓ Standard deviation $SD = \sqrt{M''(0) - \{M'(0)\}^2}$

Uniform (Discrete) distribution

Let us consider a distribution having m discrete outcomes with equal probability, and then the probability distribution of getting any of the outcomes is called the Uniform distribution of discrete type. The *pmf* of the Uniform distribution is defined as

$$f(x) = \frac{1}{m} ; x = 1, 2, 3, \dots, m$$

The *cdf* of the Uniform distribution is defined as $F(x) = P(X \leq x) = \begin{cases} 0 & ; x < 1 \\ \frac{x}{m} & ; 1 \leq x < m \\ 1 & ; x \geq m \end{cases}$

$$\text{Here, } \sum_{x \in S} f(x) = \sum_{x=1}^m \frac{1}{m} = \frac{1}{m} m = 1$$

$$\checkmark \quad E[X] = \sum_{x \in S} x f(x) = \sum_{x=1}^m x \frac{1}{m} = \frac{1}{m} \frac{m(m+1)}{2} = \frac{m+1}{2}$$

$$\checkmark \quad E[X^2] = \sum_{x \in S} x^2 f(x) = \sum_{x=1}^m x^2 \frac{1}{m} = \frac{1}{m} \frac{m(m+1)(2m+1)}{6} = \frac{(m+1)(2m+1)}{6}$$

$$\Rightarrow \text{Mean } \mu = E[X] = \frac{m+1}{2}$$

$$\Rightarrow \text{Variance } \sigma^2 = E[X^2] - \{E[X]\}^2 = \frac{(m+1)(2m+1)}{6} - \left\{\frac{m+1}{2}\right\}^2 = \frac{m^2-1}{12}$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{\frac{m^2-1}{12}}$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x) = \sum_{x=1}^m e^{tx} \frac{1}{m} = \sum_{x=1}^m \frac{e^{tx}}{m}$$

$$\text{➤ } M(0) = \sum_{x=1}^m \frac{1}{m} = 1$$

Hypergeometric distribution

Let us consider a distribution having N discrete outcomes that can be divided into two well-defined categories with N_1 and N_2 outcomes $N = N_1 + N_2$, respectively. The probability distribution of getting $x \leq N_1$ outcomes from the first category and the rest $n - x \leq N_2$ outcomes from the second category for n selected outcomes is called the Hypergeometric distribution. The *pmf* of the Hypergeometric distribution is defined as

$$f(x) = \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} ; x \leq n$$

The *cdf* of the Hypergeometric distribution is defined as $F(x) = P(X \leq x) = \sum_{x=0}^{x_t} \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}}$

$$\text{Here, } \sum_{x \in S} f(x) = \sum_{x \in S} \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = 1$$

$$\checkmark \quad E[X] = \sum_{x \in S} x f(x) = \sum_{x \in S} x \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = n \left(\frac{N_1}{N} \right)$$

$$\checkmark \quad E[X^2] = \sum_{x \in S} x^2 f(x) = \sum_{x \in S} x^2 \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = \frac{n(n-1)N_1(N_1-1)}{N(N-1)} + n \left(\frac{N_1}{N} \right)$$

$$\Rightarrow \text{Mean } \mu = E[X] = n \left(\frac{N_1}{N} \right)$$

$$\Rightarrow \text{Variance } \sigma^2 = E[X^2] - \{E[X]\}^2 = \frac{n(n-1)N_1(N_1-1)}{N(N-1)} + n \left(\frac{N_1}{N} \right) - \left\{ n \left(\frac{N_1}{N} \right) \right\}^2 = n \left(\frac{N_1}{N} \right) \left(\frac{N_2}{N} \right) \left(\frac{N-n}{N-1} \right)$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{n \left(\frac{N_1}{N} \right) \left(\frac{N_2}{N} \right) \left(\frac{N-n}{N-1} \right)}$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x) = \sum_{x \in S} e^{tx} \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}}$$

$$\triangleright \quad M(0) = \sum_{x \in S} \frac{\binom{N_1}{x} \binom{N_2}{n-x}}{\binom{N}{n}} = 1$$

Geometric distribution

Let us consider p and q as the probability of success and failure of any trial with $p + q = 1$. The trial will be terminated at the maiden success, and then the probability distribution of getting success in x -th trial in successive trials is called the Geometric distribution. The *pmf* of the Geometric distribution is defined as

$$f(x) = q^{x-1}p ; x = 1, 2, 3, \dots \dots \dots$$

The *cdf* of the geometric distribution is defined as $F(x) = P(X \leq x) = \sum_{x=1}^x q^{x-1}p$

$$\text{Here, } \sum_{x \in S} f(x) = \sum_{x=1}^{\infty} q^{x-1}p = p + qp + q^2p + q^3p + \dots \dots \dots = \frac{p}{1-q} = 1$$

$$\Rightarrow \text{Mean } \mu = E[X] = \sum_{x \in S} xf(x) = \sum_{x=1}^{\infty} xq^{x-1}p = \frac{1}{p} \text{ [Considering } p = 1 - q \text{]}$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x) = \sum_{x=1}^{\infty} e^{tx} q^{x-1}p = \frac{pe^t}{1-qe^t}$$

$$\triangleright M(0) = \frac{p}{1-q} = 1$$

$$\triangleright M'(t) = \frac{pe^t}{(1-qe^t)^2}$$

$$\triangleright M''(t) = \frac{pe^t(1+qe^t)}{(1-qe^t)^3}$$

$$\Rightarrow \text{Mean } \mu = M'(0) = \frac{1}{p}$$

$$\Rightarrow \text{Variance } \sigma^2 = M''(0) - \{M'(0)\}^2 = \frac{1+q}{p^2} - \left\{\frac{1}{p}\right\}^2 = \frac{q}{p^2}$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{\frac{q}{p^2}}$$

Binomial distribution

Let us consider p and q as the probability of success and failure of n independent trials with $p + q = 1$. The value of p and q remains the same on each trial, and then the probability distribution of getting x number of successes in n trials is called the Binomial distribution. The *pmf* of the Binomial distribution is defined as

$$f(x) = {}^nC_x p^x q^{n-x} ; x = 0, 1, 2, 3, \dots, n$$

The *cdf* of the Binomial distribution is defined as $F(x) = P(X \leq x) = \sum_{x=0}^x {}^nC_x p^x q^{n-x}$

Here, $\sum_{x \in S} f(x) = \sum_{x=0}^n {}^nC_x p^x q^{n-x} = (q + p)^n = 1$

The Binomial distribution can be represented as $b(n, p)$.

$\Rightarrow$ Mean $\mu = E[X] = \sum_{x \in S} x f(x) = \sum_{x=0}^n x {}^nC_x p^x q^{n-x} = np$ [Considering $p = 1 - q$]

$\Rightarrow$ Moment generating function $M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x) = \sum_{x=0}^n e^{tx} {}^nC_x p^x q^{n-x} = (q + pe^t)^n$

$$\triangleright M(0) = (q + p)^n = 1$$

$$\triangleright M'(t) = n(q + pe^t)^{n-1} pe^t$$

$$\triangleright M''(t) = n(n-1)(q + pe^t)^{n-2} (pe^t)^2 + n(q + pe^t)^{n-1} pe^t$$

$\Rightarrow$ Mean $\mu = M'(0) = np$

$\Rightarrow$ Variance $\sigma^2 = M''(0) - \{M'(0)\}^2 = n(n-1)p^2 + np - \{np\}^2 = np(1-p) = npq$

$\Rightarrow$ Standard deviation $SD = \sqrt{npq}$

Poisson distribution

Let us consider p as the probability of success of n trials, with $n \rightarrow \infty$ implies $p \rightarrow 0$. The probability distribution of getting x number of successes is called the Poisson distribution. Then, a finite constant $\lambda > 0$ is known as the Poisson parameter and can be attained as $\lambda = np$. The *pmf* of the Poisson distribution is defined as

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} ; x = 0, 1, 2, 3, \dots \dots \dots$$

The *cdf* of the Poisson distribution is defined as $F(x) = P(X \leq x) = \sum_{x=0}^x \frac{\lambda^x e^{-\lambda}}{x!}$

$$\text{Here, } \sum_{x \in S} f(x) = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1$$

$$\Rightarrow \text{Mean } \mu = E[X] = \sum_{x \in S} x f(x) = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} x \frac{\lambda^x}{x!} = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \sum_{x \in S} e^{tx} f(x) = \sum_{x=0}^{\infty} e^{tx} \frac{\lambda^x e^{-\lambda}}{x!} = e^{\lambda(e^t - 1)}$$

$$\triangleright M(0) = e^{\lambda(1-1)} = 1$$

$$\triangleright M'(t) = \lambda e^t e^{\lambda(e^t - 1)}$$

$$\triangleright M''(t) = (\lambda e^t)^2 e^{\lambda(e^t - 1)} + \lambda e^t e^{\lambda(e^t - 1)}$$

$$\Rightarrow \text{Mean } \mu = M'(0) = \lambda$$

$$\Rightarrow \text{Variance } \sigma^2 = M''(0) - \{M'(0)\}^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{\lambda}$$

Uniform (Continuous) distribution

Let us consider a distribution having constant probability within the interval $[a, b]$, and then the distribution of the probability in that interval is called the Uniform distribution of continuous type. The *pdf* of the Uniform distribution is defined as

$$f(x) = \frac{1}{b-a} ; a \leq x \leq b$$

The *cdf* of the Uniform distribution is defined as $F(x) = P(X \leq x) = \begin{cases} 0, & x < a \\ \int_a^x \frac{1}{b-a} dw = \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$

$$\text{Here, } \int_{-\infty}^{\infty} f(x) dx = \int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} (b-a) = 1$$

The Uniform distribution can be represented as $U(a, b)$.

$$\checkmark E[X] = \int_{-\infty}^{\infty} xf(x) dx = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2}$$

$$\checkmark E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2+ab+b^2}{3}$$

$$\Rightarrow \text{Mean } \mu = E[X] = \frac{a+b}{2}$$

$$\Rightarrow \text{Variance } \sigma^2 = E[X^2] - \{E[X]\}^2 = \frac{a^2+ab+b^2}{3} - \left\{\frac{a+b}{2}\right\}^2 = \frac{(b-a)^2}{12}$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{\frac{(b-a)^2}{12}}$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_a^b \frac{e^{tx}}{b-a} dx = \begin{cases} \frac{e^{bt}-e^{at}}{t(b-a)}, & t \neq 0 \\ 1, & t = 0 \end{cases}$$

$$\triangleright M(0) = 1$$

Exponential distribution

Let us consider a distribution having average waiting time $\theta = \frac{1}{\lambda} > 0$ for the first occurrence defined in $[0, \infty)$, and then the distribution of the probability in that interval is called the Exponential distribution. The *pdf* of the Exponential distribution is defined as

$$f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}} ; 0 \leq x < \infty$$

The *cdf* of the Exponential distribution is defined as

$$\begin{aligned} \text{➤ } F(x) &= P(X \leq x) = \int_0^x \frac{1}{\theta} e^{-\frac{w}{\theta}} dw = 1 - e^{-\frac{x}{\theta}} ; 0 \leq x < \infty \\ \text{➤ } P(X > x) &= 1 - P(X \leq x) = 1 - F(x) = e^{-\frac{x}{\theta}} ; 0 \leq x < \infty \end{aligned}$$

$$\text{Here, } \int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = 1 - 0 = 1$$

$$\Rightarrow \text{Median } m = \theta \ln 2$$

$$\Rightarrow \text{Moment generating function } M(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^{\infty} e^{tx} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \frac{1}{1-\theta t}$$

$$\text{➤ } M(0) = \frac{1}{1-0} = 1$$

$$\text{➤ } M'(t) = \frac{\theta}{(1-\theta t)^2}$$

$$\text{➤ } M''(t) = \frac{2\theta^2}{(1-\theta t)^2}$$

$$\Rightarrow \text{Mean } \mu = M'(0) = \theta$$

$$\Rightarrow \text{Variance } \sigma^2 = M''(0) - \{M'(0)\}^2 = 2\theta^2 - \theta^2 = \theta^2$$

$$\Rightarrow \text{Standard deviation } SD = \theta$$

Gamma distribution

Let us consider a distribution having average waiting time $\theta = \frac{1}{\lambda} > 0$ defined in $[0, \infty)$, and then the distribution of the probability waiting for the α -th occurrence in that interval is called the Gamma distribution. The *pdf* of the Gamma distribution is defined as

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} ; 0 \leq x < \infty$$

The *cdf* of the Gamma distribution is defined as $F(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} \int_0^x x^{\alpha-1} e^{-\frac{x}{\theta}} dx ; 0 \leq x_t < \infty$

$$\text{Here, } \int_{-\infty}^{\infty} f(x) dx = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \frac{1}{\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} dx = \frac{1}{\Gamma(\alpha)} \Gamma(\alpha) = 1$$

$$\begin{aligned} \Rightarrow \text{Moment generating function } M(t) &= E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_0^{\infty} e^{tx} \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} dx = \frac{1}{(1-\theta t)^\alpha} \end{aligned}$$

$$\text{➤ } M(0) = \frac{1}{(1-0)^\alpha} = 1$$

$$\text{➤ } M'(t) = \frac{\alpha\theta}{(1-\theta t)^{\alpha+1}}$$

$$\text{➤ } M''(t) = \frac{\alpha(\alpha+1)\theta^2}{(1-\theta t)^{\alpha+2}}$$

$$\Rightarrow \text{Mean } \mu = M'(0) = \alpha\theta$$

$$\Rightarrow \text{Variance } \sigma^2 = M''(0) - \{M'(0)\}^2 = \alpha(\alpha+1)\theta^2 - \alpha^2\theta^2 = \alpha\theta^2$$

$$\Rightarrow \text{Standard deviation } SD = \sqrt{\alpha\theta^2}$$

Normal distribution

Let us consider a symmetric distribution (Mean, Median, Mode coincide) having mean $-\infty < \mu < \infty$ and variance σ^2 ; $0 < \sigma < \infty$, defined in $(-\infty, \infty)$ is called the Normal distribution. The *pdf* of the Normal distribution is defined as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$$

For the standardized variable $z = \frac{x-\mu}{\sigma}$ *cdf* of the Normal distribution is defined as

$$\phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} dw; -\infty < z < \infty$$

$$\text{Here, } \int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

The Normal distribution can be represented as $N(\mu, \sigma^2)$.

$$\begin{aligned} \Rightarrow \text{Moment generating function } M(t) &= E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \end{aligned}$$

$$\begin{aligned} \Rightarrow M(0) &= e^0 = 1 \\ \Rightarrow M'(t) &= (\mu + \sigma^2 t) e^{\mu t + \frac{1}{2}\sigma^2 t^2} \\ \Rightarrow M''(t) &= [(\mu + \sigma^2 t)^2 + \sigma^2] e^{\mu t + \frac{1}{2}\sigma^2 t^2} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Mean } \mu &= M'(0) = \mu \\ \Rightarrow \text{Variance } \sigma^2 &= M''(0) - \{M'(0)\}^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2 \\ \Rightarrow \text{Standard deviation } SD &= \sqrt{\sigma^2} \end{aligned}$$

- ❖ $P(Z \leq z) = \phi(z)$
- ❖ $P(z_1 \leq Z \leq z_2) = \phi(z_2) - \phi(z_1)$
- ❖ $P(-z_2 \leq Z \leq -z_1) = P(z_1 \leq Z \leq z_2) = \phi(z_2) - \phi(z_1)$
- ❖ $P(Z \geq z) = 1 - P(Z \leq z) = 1 - \phi(z)$
- ❖ $P(Z \geq -z) = P(Z \leq z) = \phi(z)$
- ❖ $P(Z \leq -z) = P(Z \geq z) = 1 - \phi(z)$
- ❖ $P(|Z| \leq z) = P(-z \leq Z \leq z) = \phi(z) - \phi(-z) = \phi(z) - [1 - \phi(z)] = 2\phi(z) - 1$
- ❖ $P(|Z| \geq z) = P(Z \leq -z) + P(Z \geq z) = [1 - \phi(z)] + [1 - \phi(z)] = 2 - 2\phi(z)$
- ❖ $P(Z \leq 0) = \phi(0) = 0.5$
- ❖ $P(Z \geq 0) = P(Z \leq 0) = \phi(0) = 0.5$
- ❖ $P(Z = z) = 0$ or undefined